10 The isotope americium-241 is represented by

$$^{241}_{95}\text{Am}.$$

This isotope decays by an α-emission to an isotope of neptunium (Np).

(a) Complete the nuclide equation for this decay.

$$^{241}_{95}\text{Am} \rightarrow \overset{\cdots\cdots}{\underset{\cdots\cdots}{\quad}}\text{Np} + \overset{\cdots\cdots}{\underset{\cdots\cdots}{\quad}}\alpha$$

[3]

(b) Fig. 10.1 shows a simple diagram of a smoke detector. The smoke detector contains a small sample of americium-241. This isotope ionises the air between the metal plates in the detector.

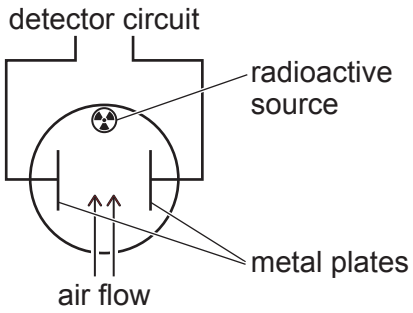


Fig. 10.1

(i) Describe how the americium-241 ionises air.

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..... [3]

(ii) Suggest and explain **two** reasons why smoke detectors use an isotope that emits α-particles rather than an isotope that emits γ-radiation.

1. .....

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2. .....

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[2]

[Total: 8]